Name- ISHA SHAW
PRN NO- 26030242023

AI-POWERED TOOLS, EXPERIENTIAL LEARNING

InnoVista Business Summit 2026

Event analytics and check-in tool: project report

AI-use disclosure: This project was built with Claude (Anthropic) as an AI assistant for data inspection, code, workbook formulas and drafting; every figure was cross-checked against the original workbook, and every prompt is recorded in the AI Use Log.

1. Objective

The goal was to build a registration, attendance and NPS dashboard and a check-in tool for the 3 January 2026 event (venue capacity 450), using only the uploaded workbook, and to turn the results into a marketing plan for next year.

- Questions answered: which channels bring attendees who actually show up, and how a ₹1 lakh budget (an assumption) should be used to fill 450 seats with attendees.
- Definitions fixed in advance: attendance rate = checked_in "Yes" ÷ registrations; no-show rate = "No" ÷ registrations; NPS = % promoters (9 to 10) minus % detractors (0 to 6) among valid scores only, always shown with n.

2. Data and method

- **Data:** the Registrations sheet has 500 rows and 12 columns. registration_id is the unique key (500 distinct), while attendee_name is not (62 names repeat across 132 rows). There is no Gold Labels sheet, so accuracy was checked against the data itself.
- **Quality checks (Phase 1):** 0 duplicate IDs; fees match attendee type and early-bird status on all 500 rows; all dates fall between 1 Nov and 30 Dec 2025; the early-bird flag matches the 1 Dec cut-off exactly. NPS is blank for all 67 no-shows and for 178 attendees, leaving n = 255.
- **Method:** Excel formulas on the sheet, an independent , Wilson 95% intervals, chi-square and Fisher tests with Holm correction, and a DATA / ASSUMPTION tag on every planning number. The workbook has no cost or spend data.
- **Headline baseline:** 500 registrations, 433 check-ins (86.6%), 67 no-shows (13.4%), NPS +23.9 (n = 255; average score 8.00), revenue ₹4,76,481.40 if every registrant paid.

3. Dashboard and check-in tool

Both pages are delivered as one standalone HTML file that opens by double-click, with all 500 rows embedded and no internet needed.

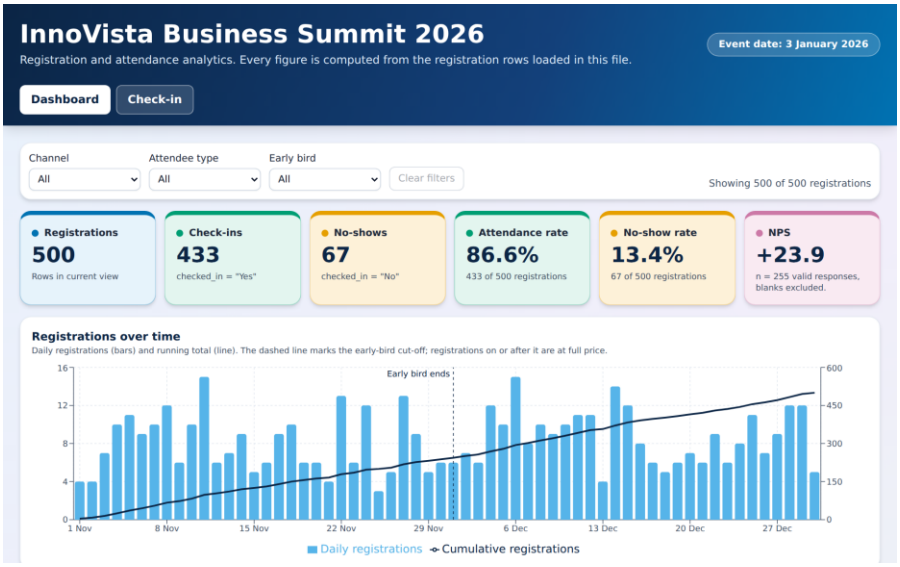


Figure 1. Dashboard: six KPI cards, filters (channel, attendee type, early bird) and the registration trend.

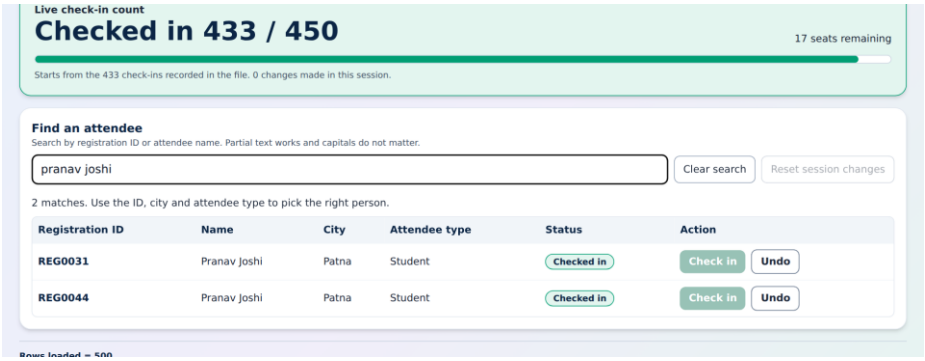


Figure 2. Check-in page: live "Checked in 433 / 450" counter and a duplicate-name search that lists every match.

- **Dashboard:** KPI cards for registrations, check-ins, no-shows, attendance, no-show and NPS with n; trend chart with the 1 Dec early-bird marker; channel charts and an NPS split; every figure computed in code from the rows.
- **Check-in:** one search box for ID or name (partial, any capitalisation, any word order); "Pranav Joshi" returns REG0031 and REG0044; status is tracked per registration_id, so nobody is counted twice; Check in and Undo buttons; stops at 450.
- **Improvement rounds (each with evidence):** (1) filters and search were lost when switching pages, and the dashboard showed 433 while the counter showed 434, so state is now kept and the dashboard can optionally include session check-ins; (2) the channel table gained share of check-ins and 95% intervals because attendance rates alone hide uncertainty; (3) an NPS small-sample warning, average score and response rate were added because one filtered view had only n = 10; (4) search now accepts words in any order because "joshi pranav" returned 0 results. A visual styling round (palette, header band) and a display fix (a missing "WhatsApp group" axis label found in a real browser) were also made.

4. Channel analysis

Channel	Reg.	Checked in	No-show	Attendance (95% CI)	Share of check-ins	NPS (n)	Avg fee ₹
College mailer	92	81	11	88.0% (79.8 to 93.2)	18.7%	+2.2 (45)	1,003.5
WhatsApp group	88	78	10	88.6% (80.3 to 93.7)	18.0%	+12.2 (41)	773.1
Friend referral	87	76	11	87.4% (78.8 to 92.8)	17.6%	+38.1 (42)	1,054.3
Website	79	69	10	87.3% (78.2 to 93.0)	15.9%	+36.6 (41)	996.6
Instagram	77	69	8	89.6% (80.8 to 94.6)	15.9%	+26.2 (42)	912.1
LinkedIn	77	60	17	77.9% (67.5 to 85.7)	13.9%	+29.5 (44)	979.7
All channels	**500**	**433**	**67**	**86.6%**	**100%**	**+23.9 (255)**	**952.96**

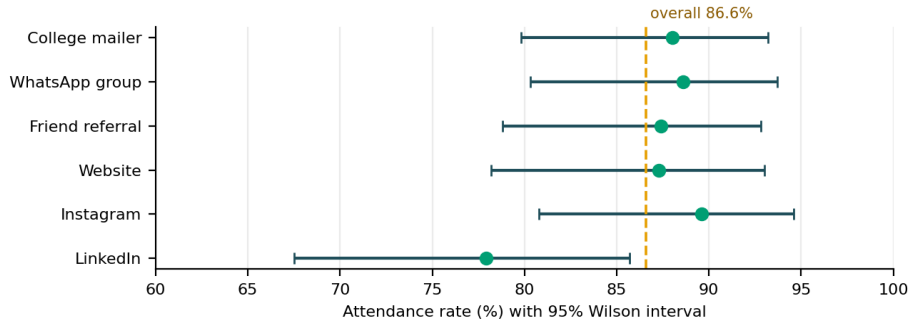


Figure 3. Attendance rate by channel with 95% intervals; the dashed line is the overall rate.

- **Not statistically meaningful:** attendance differences between channels are not significant (chi-square p = 0.291; permutation test p = 0.290; effect size 0.111). No pairwise comparison survives Holm correction. LinkedIn alone looks low (raw p = 0.027) but its Holm-adjusted p is 0.164, and with about 77 registrations per channel a 10-point gap would be detected only about 37% of the time.
- **Volume is not strength:** the College mailer leads on registrations (92) and share of check-ins (18.7%) but is third on attendance rate; Instagram has the highest rate (89.6%) with a smaller share (15.9%). The top five channels sit within 2.3 points of each other.
- **Attendee-type mix:** Faculty attend at 51.4% (19 of 37; 95% CI 35.9% to 66.6%) against 88.8% Students, 88.1% Professionals and 95.2% Founders (chi-square p = 9.9e-10; Fisher Faculty vs others p = 5.4e-08). Faculty are 7.4% of registrants but 26.9% of no-shows. Type mix explains about 2.7 of LinkedIn's 8.7-point gap; after adjusting for type its effect weakens (p = 0.087).
- **NPS by channel:** differences are not significant (p = 0.536), with only 41 to 45 responses per channel. This is a pattern, not proof of causation: a channel with a higher NPS does not show that the channel produces happier attendees.

5. Recommendation (quantified)

Tags: [DATA] = from the workbook; [ASSUMPTION] = anything else, including the ₹1 lakh budget.

- **Registrations needed:** 450 attendees at the overall 86.6% need about 520 registrations [DATA], or 524 when each channel is rounded up, against 500 this year. Planning at the bottom of each interval would need 582 [DATA].

Budget line	Registrations needed [DATA]	Budget ₹ [ASSUMPTION]	% of ₹1 lakh [ASSUMPTION]
College mailer	96	13,800	13.8%
WhatsApp group	92	13,200	13.2%
Friend referral	91	13,050	13.05%
Website	83	11,850	11.85%
Instagram	81	11,550	11.55%
LinkedIn	81	11,550	11.55%
No-show programme	n/a	15,000	15%
Reserve (released at the T-33 checkpoint)	n/a	10,000	10%
Total	**524**	**1,00,000**	**100%**

- **How the split was made:** ₹75,000 is divided in proportion to registrations needed. Because each channel keeps its share of this year's attendees, this is a straight scale-up (this year's registrations × 450 ÷ 433), not a ranking of channel quality. The data cannot separate the channels, so no money is moved toward or away from any channel.
- **Timeline:** registration ran 63 days before the event; 238 registrations (47.6%) came before the 1 Dec early-bird deadline, but there was no rush: 24 to 30 Nov had 53 registrations against 64 in the first week of December, and the busiest week was 6 to 12 Dec (74). The early-bird discount cost ₹51,055.60 [DATA]. Plan the same T-63 / T-33 / T-4 structure [ASSUMPTION], keep pushing through December, and run reminders at T-7, T-2 and T-1.
- **Faculty action:** confirm-or-release: request confirmation by T-14, remind at T-7 and T-1, and release unconfirmed seats to a standby list at T-3 [ASSUMPTION]. A 75% Faculty attendance target [ASSUMPTION] adds about 8.75 attendees on the same 37 registrations, lowers registrations needed to 510, and is funded inside the existing ₹15,000 no-show line (about ₹4,030, or ₹109 per Faculty registration); the allocation does not change.
- **Three risks:** (1) attendance below plan: each 1-point drop needs about 6 more registrations [DATA]; keep a standby list and the reserve. (2) Over-capacity: 524 registrations at 86.6% gives about 454 attendees [DATA]; stop or waitlist once expected attendance reaches 450. (3) Thin evidence and no cost data: track spend by channel from day one and use the reserve at T-33.

6. Verification results

Check	Result
App figures vs independent recalculation from the xlsx (full data plus Instagram-only and Student-only views)	105 of 105 match
Same comparison after the improvement rounds (full data and four filtered views)	100 of 100 match
Workbook: Baseline Validation / Channel Analysis / Marketing Plan	Workbook: Baseline Validation / Channel Analysis / Marketing Plan — 19 of 20, 127 of 134, and 18 of 18 comparisons currently MATCH; 1 Baseline Validation mismatch and 7 Channel Analysis mismatches.
Workbook: segment and NPS sections (Sections D to G)	105 comparisons all MATCH
Formula errors in the workbook; changes to the original five sheets	0; 0
Standalone HTML opened in a real browser (headless Chromium, file://)	Same key figures; no network requests; no errors

7. Limitations

- There is no cost or spend data, so there is no ROI or cost per attendee; the budget split is a recommendation, not an optimisation.
- Revenue assumes every registrant paid; the workbook has no payment or refund field.
- One year of data, about 77 to 92 registrations and 41 to 45 NPS responses per channel, so channel differences cannot be separated from chance. Only 51% of registrants and 59% of attendees answered the NPS question.
- heard_about_via is a single self-reported answer, so all findings are associations, not causation. Several findings, such as LinkedIn Professionals at 73.3% (22 of 30), came from subgroup checks made after seeing the data and are hypotheses to test.
- Check-ins made in the tool are not saved after the page is reloaded, and the dashboard was tested in Chromium only.

8. Responsible-AI notes

- AI produced the code, formulas and drafts; I did not accept any number without a separate check. The event and records are fictional, and no personal data was used.
- Verification caught AI errors: NPS was first rounded to about +24.0 (the exact value is 23.92); a peak-day check wrongly assumed which tied date Excel would return; the Excel interval formulas used a rounded constant; the workbook was first delivered with 7 MISMATCH cells; and a missing axis label appeared only in a real browser.
- Safeguards used: fixed definitions, DATA vs ASSUMPTION tags, no causal claims, small-sample warnings, and an explicit "not significant" statement where the data could not separate groups.
- The AI Use Log records every prompt, including a rejected output (removing the Undo button, which I reversed) and a partly kept output (the workbook delivered with the 7 MISMATCH cells).

9. Personal reflection

The main lesson was that a clean-looking answer is not the same as a checked one. Several of the AI's outputs looked finished but contained small errors that only an independent recalculation exposed, such as the rounded 1.96 constant and the first-delivered workbook with 7 MISMATCH cells. Asking for exact Excel formulas and independent values early made those errors easy to find.

I also learned that "not significant" is a result, not a failure: the channel differences that looked interesting (LinkedIn, the NPS ranking) could not be separated from chance, so the honest plan keeps the channel split proportional and puts the clearest action on Faculty, where the evidence is strong. One decision I made myself was asking for the Undo button to be removed and then restored. Next time I would ask for cost data before planning a budget, and I would test whether the Faculty confirmation step actually lifts attendance.

10. Conclusion

The dashboard and check-in tool work on all 500 registrations, with every figure checked against the original data: 433 attended (86.6%) and NPS was +23.9 (n = 255). Attendance did not differ significantly between channels ($p = 0.291$), so the budget split stays proportional. The clearest finding is that Faculty attended only 51.4% of the time, so the plan adds a Faculty confirmation step. Filling 450 seats next year needs about 520 registrations. The budget split rests on assumptions because the data has no cost figures, and the results show association, not cause.